Lunar lander mast deployment failure

"our idea is to develop a compact inflatable composite carbon fiber mast with communication hardware, flexible solar panel that is attached to the inflated mast. these parts fitted inside a cylindrical lander with three legs that has low reaction force drill system to anchor itself on the moon surface and give rigid support . what are the alternative ways if the tower is failed to land properly?"
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Contingency Architectures for a Compact Lunar Mast-and-Solar Lander at a Glance

Key Findings Table

Topic	Key finding	Relevance to your concept	Citations
Best overall contingency approach	The strongest pattern is **graceful degradation**: do not make the mast mission-critical; preserve minimum communications and power on the chassis alone.	Your inflatable mast should be a performance upgrade, not the only way to survive landing.	[1] [2] [3] [4] [5] [6] [7]
Best structural fallback mast	**Rollable/self-erecting composite booms with guy wires or spreader bars** are better characterized than inflatable mast alternatives for load/deflection tradeoffs.	Strong candidate if you want a second mast option or replace the inflatable baseline.	[8] [9] [10] [11] [12]
Backup communications	**Body-mounted patch antennas** can maintain low-bandwidth command/telemetry without pointing; optical beacons and black-box radios can send survival/status data after partial failure.	Allows the lander to stay reachable even if the tower is unusable.	[1] [2] [3] [13] [14]
Relay continuity	**Orbital, Lagrange-point, and networked relay architectures** can preserve coverage when a direct-to-Earth mast cannot be used.	Especially important for polar or far-side sites.	[4] [5] [15]

Topic	Key finding	Relevance to your concept	Citations
Backup power	**Body-mounted, fold-out, blanket, ground-deployed, tent-shaped, and elevated arrays** offer non-mast power options; elevated arrays were mass-favorable in one lunar trade space.	Power recovery is more mature than mast recovery.	[6] [7] [16] [17] [18]
Landing resilience	**Touchdown stability strongly determines whether any tall deployable remains usable**; active or semi-active legs improve slope tolerance and reduce tip-over risk.	Investing in landing/leveling may improve total mission robustness more than mast optimization alone.	[19] [20] [21] [22] [23] [24]
Anchoring	Helical anchors, modular footings, and regolith pads appear feasible for lunar load transfer, but **mast-specific support capacity is not well quantified**.	Your drill-anchor concept is plausible, but tower bending/torsion support remains a key unknown.	[25] [26] [27] [28]
Recovery after bad landing	**Self-righting, reconfigurable legs, hopping torques, and post-landing reorientation** are emerging ways to recover utility after off-nominal touchdown.	Useful if the lander arrives tilted and the mast cannot safely inflate.	[29] [30] [31] [32]

Direct Answer

If the tower fails to land properly, the best alternatives are not just “another tower,” but a **fault-tolerant architecture** with several fallback modes. The most practical stack is: **(1)** survive touchdown with compliant or semi-active legs and a low center of gravity, **(2)** assess tilt, sink, and mast condition, **(3)** self-right or re-level if possible, **(4)** if the mast is unusable, switch to **body-mounted backup antennas**, **(5)** deploy **non-mast solar** such as body panels, fold-out wings, or ground blankets, **(6)** anchor only the chassis, and **(7)** route higher-performance communications through relay assets rather than relying on mast height alone [1] [2] [4] [6] [7] [15] [19] [21] [22] [29] [30].

For a **structural fallback**, the best-supported alternative in the supplied evidence is a **rollable composite boom with guy wires or spreader bars**, because these systems have clearer analyses for deflection control and tip-mass support than the inflatable mast literature in this set [8] [9] [10] [12]. If you do keep the inflatable concept, treat it as a **jettisonable or bypassable performance tower**, while all essential avionics, minimum solar area, and low-rate communications remain functional from the lander body [1] [7] [18].

Confidence level: High - the fallback architecture is supported across multiple independent themes: communications redundancy, alternative solar layouts, landing stability, and post-landing recovery [1] [5] [6] [19] [30].

Study Scope

This report synthesizes the supplied evidence on: deployable masts and booms, backup communications, lunar relay architectures, alternative solar deployment, touchdown stability, anchoring in regolith, and post-landing recovery concepts, largely spanning studies from roughly 2000-2025 and including simulations, trade studies, subsystem experiments, and mission concept papers [33] [34] [35] [36] [37] [38].

Assumptions & Limitations

The evidence base is fragmented across subsystem silos, with few end-to-end studies covering the entire chain from touchdown through mast deployment, anchoring, and contingency operations [33] [34] [35] [36] [37] [38]. The supplied findings do **not** provide validated mast-specific lunar anchor capacity for your exact three-leg cylindrical lander with low-reaction-force drilling, nor a direct quantitative reliability comparison between inflatable rigidizable masts and rollable composite booms for lunar communications payloads [8] [9] [10] [11] [25] [26] [27] [28].

Suggested Further Research

- Run an integrated simulation that couples **touchdown tilt, leg sink, mast inflation/erection, and drilling-anchor loads** in one model [20] [25] [39].
- Quantify whether a **rollable composite boom** or **inflatable rigidizable mast** gives better survivability for your payload mass and allowable deflection [8] [9] [10] [11].
- Test a **dual-mode flexible solar system** that can attach to the mast nominally but also deploy onto the ground or side petals after mast failure [7] [17].
- Add and validate a **minimum viable survival mode**: body antenna, low-rate beacon, and chassis-mounted solar [1] [2] [18].
- Develop a **remote release or jettison path** for a damaged mast, since this is a documented gap in the retrieved literature [40] [41] [42].

Full Report

1. Foundational context

Your concept combines five tightly coupled subsystems: a compact inflatable composite carbon-fiber mast, communications hardware, flexible solar attached to that mast, a cylindrical three-leg lander, and a low-reaction-force drill/anchor intended to create a rigid support on the Moon. The retrieved literature suggests that this is not just a mast problem; it is a **landing dynamics plus deployable structure plus power/communications redundancy** problem [11] [43] [44] [45].

Three background ideas are especially important.

First, **tall deployables are highly sensitive to base condition**. Studies of lunar touchdown repeatedly show that attitude, horizontal velocity, slope, sink, regolith interaction, and leg compliance strongly affect post-landing stability and overturn risk [20] [23] [24]. That means a good mast can still fail operationally if the chassis lands tilted or unstable [19] [21].

Second, the literature favors **graceful degradation** over single-point dependency. Across communications and power studies, the most resilient architectures maintain some capability without the primary mast or high-gain antenna, using body antennas, beacons, relays, and alternative array layouts [1] [2] [5] [6] [7].

Third, there is an evidence asymmetry: **composite boom towers are better quantified than inflatable mast towers** in the supplied set. Inflatable systems remain attractive for mass and packaging, but the retrieved findings emphasize viscoelastic growth, pressure loss, and weaker direct load/deflection comparisons than boom-based studies provide [8] [9] [10] [11] [39].

Confidence level: High - these themes recur across multiple independent evidence streams [1] [6] [19] [39].

2. Problem framing and failure scenarios

2.1 Integrated system concept and mission role

The proposed system is effectively a compact lunar surface utility node: the mast elevates communications hardware and supports attached flexible solar, while the lander body provides storage volume, touchdown structure, and the base for anchoring into regolith [11] [43] [44] [45]. Similar retrieved concepts include deployable mast-plus-solar systems for lunar environments and self-erecting surface towers intended to support communications or sensors above a lander deck [40] [46].

Taken together, these findings suggest your mission utility depends on three things being true at once: **the lander must remain stable, the mast must deploy correctly, and the mast-chassis interface must carry lateral loads after anchoring** [19] [25] [39]. That coupling is the main reason to plan alternatives early.

2.2 Failure modes across descent, touchdown, and deployment

The supplied evidence identifies several relevant off-nominal modes:

- **Touchdown instability** from horizontal velocity, slope, yaw, and leg/regolith compliance, potentially leading to overturning or unsafe tilt [20] [23] [24].
- **Passive gear insufficiency** on steep slopes; one study reported passive gear could not safely land on a 20° slope, while active damping prevented tip-over in the modeled case [19].
- **Buckling, curling, or incomplete deployment** in thin-shell or rollable boom systems if geometry and root support are not controlled [47] [48] [49] [50].
- **Inflatable rigidizable mast drift** due to time-dependent diameter growth and pressure loss after rigidization from adhesive-joint viscoelastic effects [39].
- **Post-landing base inadequacy** if anchors or footings do not provide sufficient stiffness for mast bending loads, especially under uneven terrain or uncertain regolith properties [25] [27] [51].

The practical implication is that “failed to land properly” should not be treated as one event. It includes at least four distinct cases: **tilted-but-intact lander**, **upright lander with unusable mast**, **upright mast with inadequate anchoring**, and **partially damaged chassis that still has some power/comms utility** [3] [21] [25] [39].

Confidence level: High - touchdown and deployment failure modes are directly documented; end-to-end rates are not [20] [39] [47].

3. Alternative mast and tower deployment options

3.1 Self-erecting composite booms

The best-supported alternative to an inflatable mast is a **self-erecting collapsible/rollable composite boom** [12] [40] [52]. Demonstrated mechanisms in the supplied set include unspooling a collapsible carbon-fiber mast into an erect tubular state and strain-energy deployment controlled by a sprung roller and rotary damper to prevent blossoming and regulate speed [40] [52].

For lunar tower use, these systems have clearer structural characterization than inflatables. The supplied studies report appreciable static tip deflection under dead load, with a cited natural lateral deflection of about 5% of deployed boom length in one content summary, and performance constrained by maximum allowable deflection [8] [9] [10]. That is not ideal, but it is at least quantifiable.

This makes rollable booms attractive in two contingency roles:

1. **Primary replacement** for the inflatable mast if you want a more characterized erectable tower [8] [9].
2. **Short emergency mast** on the chassis, carrying only a smaller communications package rather than the full solar-attached payload [12] [40].

3.2 Guy-wire and spreader-bar stabilization

The retrieved evidence strongly supports **guy wires and spreader bars** as a way to improve lateral stability and allowable tip mass in lightweight composite towers [8] [9] [10]. The benefit appears when spreader-bar length is sufficient, and tensioned guy wires can substantially reduce tip deflection and damp oscillations quickly [10] [40].

For your concept, this suggests an important design alternative: instead of trying to recover a failed tall inflatable mast exactly as planned, use a **shorter mast plus deployed stays** anchored to the surface or to deployable outriggers on the lander [8] [9]. This pattern implies a lower-risk contingency architecture because it reduces reliance on pure cantilever stiffness.

A limitation is that the supplied lunar touchdown studies do **not** directly describe guy-wire systems for correcting landing-induced tilt; they focus more on landing gear stability than mast-stay correction hardware [33] [34] [53].

3.3 Inflatable and rigidizable variants

Inflatable/rigidizable structures remain promising because they offer low mass and compact stowage, and foundational work explicitly links rigidizable composite materials to inflatable space structures for exploration systems [43] [44]. The lunar tower material also emphasizes rapid, scalable segmented deployment for communication/navigation support [11].

However, within the supplied set, the cautionary evidence is stronger than the validation evidence for contingency use. The key reported issue is **unexpected time-dependent diameter growth and pressure loss after rigidization**, attributed to viscoelastic behavior in adhesive lap joints [39]. Also, unlike the composite boom studies, the inflatable mast content here does not provide comparable load-capacity or lateral-deflection trade studies [8] [9] [10] [11].

Therefore, if you retain the inflatable approach, the evidence supports using it as a **nominal high-performance mode**, with a separate fallback rather than expecting damaged-state recovery from the inflatable itself [1] [7] [39].

Confidence level: Medium-High - strong for relative maturity of boom alternatives, lower for quantitative inflatable-vs-boom comparison because the retrieved set lacks head-to-head reliability data [10] [11] [39].

4. Fallback communications architectures

4.1 Low-profile onboard backup antennas

A very practical fallback is to place **additive-manufactured patch antennas** on the lander body so communications do not depend on mast pointing or elevation [1] [13] [14]. In the supplied example, a 1 W RF output architecture could support up to 7.5 Mbps over 2 km, while requiring no pointing and allowing heartbeat/lost-signal messages plus command reception [1] [13] [14].

For your design, that means a minimum-viable communication mode can survive even if the mast is bent, not inflated, or deliberately abandoned [1].

4.2 After-failure beacons and black-box transmitters

The literature also supports very low-duty-cycle survival channels. A **standalone optical beacon** is described as a low-SWaP lunar-night survival terminal for status recovery via relay satellites [2]. OMOTENASHI's surface probe was designed to transmit post-arrival information including impact-acceleration history, landing state, and visible-light imagery, with about one hour of surface survival [3]. Standalone radio beacons from impacting micro-probes were also proposed to support localization and beyond-line-of-sight operations [54] [55].

These are particularly useful if your mast failure coincides with a poor attitude or low remaining power budget.

4.3 Relay-based continuity for polar and far-side sites

If the tall mast is unusable, the best way to preserve meaningful connectivity is often to **change the network architecture**, not to force local recovery [4] [5] [15]. The supplied studies emphasize Lagrange-point relays and polar-orbit relays for far-side and south-pole continuity [4] [56] [57] [58]. LunaNet-like networked approaches combine relay elements, possible surface elements, Earth systems, and DTN routing across nodes [5].

Taken together, these findings suggest that if your site geometry is challenging, you should not design around a mast as the sole path to high-performance communications. The mast can improve local link margin or visibility, but **relay dependency is often a better resilience investment** [4] [5].

Confidence level: High - multiple direct examples support body backups, beacons, and relays [1] [2] [3] [5].

5. Alternative solar power deployment strategies

5.1 Body-mounted and wing-style arrays

The simplest fallback power mode is **body-mounted solar**, which removes dependence on mast health altogether [18]. Historical progression in the supplied findings moved from body-mounted cells to extended panels and deployable twin wings as power demand increased [18]. For your concept, this supports keeping at least a **survival-level solar area on the chassis** even if the main flexible array is intended for the mast [18].

5.2 Ground-deployed and elevated arrays

The evidence suggests several alternatives if mast-mounted solar cannot be used:

- **Tent-shaped east/west arrays** with dawn/dusk illumination advantage [16].
- **Ground-deployed blankets** with separate power distribution [17] [59].
- **Elevated but separate arrays**, which were found to be the best mass option across one lunar architecture comparison, depending on illumination and height [6].

A useful inference for your system is to design the flexible solar blanket as a **dual-mode asset**: mast-attached in the nominal case, but ground-unrolled or petal-supported in the contingency case [7] [17]. This pattern is consistent with the broader finding that power contingency is more mature than tower contingency [6] [7] [40].

5.3 Flexible blanket and foldable PV systems

The strongest deployment evidence in the solar set comes from **flexible blanket and roll-out arrays**. The ROSA on-orbit demonstration reported successful deployment and retraction while verifying structural dynamics and photovoltaic viability [7]. Classic blanket arrays use flexible substrates, protective stowage, and controlled concertina deployment with tensioned booms and latches [60] [61] [62]. Reviews also describe flexible concepts as compact, robust, and better able to tolerate overload or harsh environments [63] [64] [65].

That makes a separate flexible blanket package one of the most credible fallback elements for your lander.

Confidence level: High - solar fallback options are diverse and comparatively mature in the supplied evidence [6] [7] [17].

6. Touchdown stability, anchoring, and recovery

6.1 Passive, semi-active, and active landing gear

The evidence repeatedly shows that **touchdown stability matters as much as mast design** [19] [20] [22]. One controlled landing-gear study found passive gear could not safely land on a 20° slope, while active damping reduced first-contact leg load, suppressed roll growth, and prevented tip-over [19]. Semi-active systems combining energy-absorbing honeycomb and controllable damping were evaluated across lunar-soil friction and slope cases [22]. Other work used metrics such as minimum COM-to-overturning-plane distance to track overturn risk [21].

For your concept, this argues for investing in **leg compliance, damping, and perhaps leg-length adjustment** before relying on a tall mast.

6.2 Regolith anchoring and footing concepts

Anchoring in lunar regolith appears feasible, but the evidence is not yet mast-specific. **Helical anchors** were tested for lateral resistance and uplift in lunar simulant, suggesting screw-in anchors can work on the Moon [25] [26] [66]. **Modular footers** were proposed to distribute load and stabilize on uneven terrain [27]. **Regolith-fused pads** showed meaningful structural capacity for lander footpads, though that evidence is about landing loads rather than mast overturning loads [28].

So your drill-anchor idea is reasonable in principle, but the major unresolved question is whether the resulting base stiffness is enough for a mast carrying both communications hardware and wind-free but moment-inducing flexible solar loads [25] [28].

6.3 Post-landing reconfiguration and self-righting

If the lander does not arrive in an acceptable posture, the literature supports several **recovery-by-reorientation** approaches. Small probes have used or studied anchor harpoons, ice screws, downward thrust, external springs, and hopping torques to upright after rebounds [29]. Reconfigurable or mobile legged landers can adjust leg poses after landing for hostile or uneven terrain [30] [31]. Pathfinder/MER-style petal configurations are cited as self-righting precedents in a hopper context [32].

This emerging trend suggests a practical contingency option for your three-leg chassis: **self-level first, then decide whether to inflate or abandon the mast** [19] [30] [31].

Confidence level: Medium-High - touchdown stability evidence is strong; direct lunar self-righting validation is more limited [19] [29] [30].

7. Recommended contingency architecture for your concept

Based on the supplied evidence, the most robust architecture is not a single backup device but a **tiered contingency hierarchy**.

Tier 1: Minimum viable lander without mast

Keep essential avionics, a body-mounted low-rate antenna, and minimum body solar functional with the mast fully lost [1] [18].
Confidence level: High [1] [18].

Tier 2: Touchdown tolerance and re-leveling

Use compliant or semi-active legs, low center of gravity, and if feasible reconfigurable legs or self-leveling geometry before mast deployment [19] [22] [30].
Confidence level: High [19] [22].

Tier 3: Non-mast communications survival mode

Switch to patch antennas, optical beacon, or black-box style transmitter if the mast is unusable [1] [2] [3].
Confidence level: High [1] [3].

Tier 4: Non-mast power survival mode

Deploy body panels, fold-out wings, or a flexible blanket onto the ground or side petals [7] [17] [18].
Confidence level: High [7] [17].

Tier 5: Optional secondary mast

If a mast is still desired after bad landing, prefer a shorter **rollable composite boom with guying** over trying to recover a compromised inflatable tower [8] [9] [10].
Confidence level: Medium-High [10] [11].

Tier 6: Relay-dependent high-performance mode

Use orbital or Lagrange relays for high-value links, especially at polar/far-side sites [4] [5].
Confidence level: High [4] [5].

8. Main design implications

1. **Do not couple mission survival to the mast.** The strongest cross-source pattern is that redundancy in comms and power is more robust than mast recovery itself [1] [7].
2. **Improve landing stability first.** Touchdown attitude and post-landing geometry likely dominate whether any tower can safely operate [20] [23] [24].
3. **Prefer a dual-mode solar architecture.** Flexible solar should work both on the mast and off the mast [7] [17].
4. **If you need a backup tower, use a guyed rollable composite boom.** It is better characterized than inflatable alternatives in this evidence set [8] [9] [10].

5. **Treat the drill/anchor as a chassis stabilizer first, tower foundation second.** Mast-specific overturning support remains insufficiently quantified [25] [28].

9. Conclusion

The best answer to "what are the alternative ways if the tower fails to land properly?" is: **design the mission so the tower can fail without ending the mission** [1] [5] [7]. The literature supports a layered fallback architecture consisting of touchdown-tolerant landing gear, self-leveling or self-righting capability, body-mounted backup communications, beacon/black-box survival transmitters, non-mast solar deployment, and relay-based connectivity [2] [3] [4] [17] [19] [30]. If you also want a structural backup mast, the best-supported choice in the supplied evidence is a **rollable composite boom with guy wires or spreader bars**, not a second inflatable concept [8] [9] [10] [12].

Overall confidence level: High - the architectural recommendation is strongly supported across several subsystems, though exact sizing and anchor capacity for your specific lander remain open engineering questions [1] [6] [25] [39].

References

1. Nagesh, A. (2023) LASER Relay Satellite Network for real-time mission operation on Moon, Mars and Beyond !. *Proceedings of the International Astronautical Congress, IAC*. https://www.scopus.com/pages/publications/85188011862
2. Hopkins, J., Ellis, D. (2024) Survive the Lunar Night Low-SWaP Optical Communication System. *IEEE Aerospace Conference Proceedings*. https://doi.org/10.1109/AERO58975.2024.10520943
3. Otsuki, M., Kikuchi, J., Yoshimitsu, T., Hashimoto, T. (2024) Development of hybrid landing gear for OMOTENASHI surface probe. *Acta Astronautica*. https://doi.org/10.1016/j.actaastro.2024.08.018
4. Li, H., Zhang, Q., Zhang, N. (2007) Equilibrium points in Earth-Moon system for continuous communication with far-side station and lunar orbiter. *Proceedings of SPIE - The International Society for Optical Engineering*. https://doi.org/10.1117/12.774884
5. Israel, D.J., Schier, J., Petro, A., (...), Sharma, A. (2021) LunaNet Architecture and Concept of Operations. *16th International Conference on Space Operations*. https://www.scopus.com/pages/publications/105024944772
6. Thakar, B.V., McNabb, J.T., Robertson, B.E., Mavris, D.N. (2025) Trade Studies for Evaluating Lunar Surface Power Architectures. *AIAA Science and Technology Forum and Exposition, AIAA SciTech Forum 2025*. https://doi.org/10.2514/6.2025-2192
7. Chamberlain, M.K., Kiefer, S.H., LaPointe, M., LaCorte, P. (2021) On-orbit flight testing of the Roll-Out Solar Array. *Acta Astronautica*. https://doi.org/10.1016/j.actaastro.2020.10.024
8. Daye, J.G., Lee, A.J., Fernandez, J.M. (2023) Structural Architectures for Self-Erecting Lunar Towers. *AIAA SciTech Forum and Exposition, 2023*. https://doi.org/10.2514/6.2023-0937

9. Daye, J.G., Lee, A.J., Fernandez, J.M. (2025) Structural Architectures for Imperfect Deployable Lunar Towers. *Journal of Spacecraft and Rockets*. https://doi.org/10.2514/1.A36093
10. Miller, A.S., Lordos, G., Portman, P., (...), Fernandez, J.M. (2023) Design Development of a Stable, Lightweight, Tall and Self-Deploying Lunar Tower. *IEEE Aerospace Conference Proceedings*. https://doi.org/10.1109/AERO55745.2023.10115592
11. Testi, C., Bailey, K., Shah, K., (...), Bannova, O. (2024) DESIGN OF INFLATABLE MULTI-PURPOSE TOWER FOR SUPPORT OF ROBOTIC AND CREWED LUNAR SURFACE OPERATIONS. *Proceedings of the International Astronautical Congress, IAC*. https://doi.org/10.52202/078357-0190
12. Straubel, M., Block, J., Sinapius, M., Hühne, C. (2011) Deployable composite booms for various gossamer space structures. *Collection of Technical Papers - AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference*. https://doi.org/10.2514/6.2011-2023
13. Nagesh, A. (2023) Additive Manufactured Patch Antenna Design for Lunar Surface Telemetry, Tracking and Command Link and Upstream. *Proceedings of the International Astronautical Congress, IAC*. https://www.scopus.com/pages/publications/85187995204
14. Nagesh, A. (2023) Miniaturised Design of On-board Antenna using Additive Manufacturing Techniques. *Proceedings of the International Astronautical Congress, IAC*. https://www.scopus.com/pages/publications/85188000781
15. Turnbull, E., Oleson, S., Dosa, B., (...), Fittje, J. (2023) Lunar Surface Relay – Mobile: Concept to Provide Relay Links to Surface Assets. *Accelerating Space Commerce, Exploration, and New Discovery Conference, ASCEND 2023*. https://doi.org/10.2514/6.2023-4687
16. Hickman, J.Mark, Curtis, Henry B., Landis, Geoffrey A. (1990) Design considerations for lunar based photovoltaic power systems. *Conference Record of the IEEE Photovoltaic Specialists Conference*. https://www.scopus.com/pages/publications/0025421743
17. Wrosch, M., Walmsley, N. (2015) Encapsulated photovoltaic blankets and modules for deployable solar arrays. *2015 IEEE 42nd Photovoltaic Specialist Conference, PVSC 2015*. https://doi.org/10.1109/PVSC.2015.7356140
18. Mercer, C.R. (2022) Solar array designs for deep space science missions. *Photovoltaics for Space: Key Issues, Missions and Alternative Technologies*. https://doi.org/10.1016/B978-0-12-823300-9.00006-6
19. Maeda, Takao, Otsuki, Masatsugu, Hashimoto, Tatsuaki (2019) Protection against overturning of a lunar-planetary lander using a controlled landing gear. *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*. https://doi.org/10.1177/0954410017742931
20. Cohen, B.A., Chavers, D.G., Ballard, B.W. (2012) NASA S robotic lunar lander development project. *Acta Astronautica*. https://doi.org/10.1016/j.actaastro.2012.03.025
21. Dong, Y., Ding, J., Wang, C., (...), Liu, X. (2022) Soft landing stability analysis of a Mars lander under uncertain terrain. *Chinese Journal of Aeronautics*. https://doi.org/10.1016/j.cja.2021.10.034

22. Wang, C., Nie, H., Chen, J., Lee, H.P. (2019) The design and dynamic analysis of a lunar lander with semi-active control. *Acta Astronautica*. https://doi.org/10.1016/j.actaastro.2018.12.037

23. Wan, J., Nie, H., Li, L., (...), Zeng, F. (2010) Analysis of landing performance and multi-factor effects with lunar lander. *Nanjing Hangkong Hangtian Daxue Xuebao/Journal of Nanjing University of Aeronautics and Astronautics*. https://www.scopus.com/pages/publications/77954373007

24. Zhu, W., Yang, J.-Z. (2009) Touchdown stability simulation of landing gear system for lunar lander. *Yuhang Xuebao/Journal of Astronautics*. https://doi.org/10.3873/j.issn.1000-1328.2009.05.008

25. Crabtree, J., Klosky, J.L., Gash, R., Miller, D. (2008) Helical anchors: Lateral resistance for shaping the lunar surface. *Earth and Space Conference 2008: Proceedings of the 11th Aerospace Division International Conference on Engineering, Science, Construction, and Operations in Challenging Environments*. https://doi.org/10.1061/40988(323)118

26. Klosky, J.L., Sture, S., Ko, H.-Y., Barnes, F. (1998) Helical anchors for combined anchoring and soil testing in lunar operations. *Space 1998*. https://doi.org/10.1061/40339(206)57

27. Appana, S., Formoso, O., Ochalek, M., Cheung, K. (2024) Footer Module Systems for Discrete Metamaterials Structure Construction in Lunar Environments. *IEEE Aerospace Conference Proceedings*. https://doi.org/10.1109/AERO58975.2024.10521123

28. Carter, A., Brewer, A., Garvey, R. (2024) Lunar Landing and Launch Pad Construction Using Concentrated Solar Energy to Fuse Lunar Regolith Feedstock. *Earth and Space 2024: Engineering for Extreme Environments - Proceedings of the 19th Biennial International Conference on Engineering, Science, Construction, and Operations in Challenging Environments*. https://doi.org/10.1061/9780784485736.081

29. Ge, D., Cui, P., Zhu, S. (2019) Recent development of autonomous GNC technologies for small celestial body descent and landing. *Progress in Aerospace Sciences*. https://doi.org/10.1016/j.paerosci.2019.06.002

30. Han, Y., Guo, W., Zhao, D., Li, Z. (2022) Multi-mode unified modeling and operation capability synergistic evaluation for the reconfigurable legged mobile lander. *Mechanism and Machine Theory*. https://doi.org/10.1016/j.mechmachtheory.2021.104714

31. Tian, Baolin, Liu, Jiaxing, Yang, Xuecong, (...), Deng, Zongquan (2026) Recent Progress in Aerospace Mechanisms for Reusable Launch Vehicles: A Review. *Engineering*. https://doi.org/10.1016/j.eng.2026.02.030

32. Montminy, S., Dupuis, E., Champliaud, H. (2008) Mechanical design of a hopper robot for planetary exploration using SMA as a unique source of power. *Acta Astronautica*. https://doi.org/10.1016/j.actaastro.2008.01.012

33. Ji, S., Liang, S. (2021) DEM-FEM-MBD coupling analysis of landing process of lunar lander considering landing mode and buffering mechanism. *Advances in Space Research*. https://doi.org/10.1016/j.asr.2021.03.034

34. Liu, Yuanyuan, Song, Shunguang, Li, Ming, Wang, Chunjie (2017) Landing stability analysis for lunar landers using computer simulation experiments. *International Journal of Advanced Robotic Systems*. https://doi.org/10.1177/1729881417748441

35. Delepaut, A., Giordano, P., Ventura-Traveset, J., (...), Walker, R. (2020) Use of GNSS for lunar missions and plans for lunar in-orbit development. *Advances in Space Research*. https://doi.org/10.1016/j.asr.2020.05.018

36. Ijichi, K., Oka, N., Yanagawa, H., (...), Kai, T. (2026) The concept study of the moon electric power system. *Acta Astronautica*. https://doi.org/10.1016/j.actaastro.2026.08.056

37. McKay, David S., Carter, James L., Boles, Walter W., (...), Allton, Hudith H. (1994) New lunar soil simulant. *Proceedings of the 4th International Conference on Engineering, Construction and Operations in Space*. https://www.scopus.com/pages/publications/0028022247

38. Maddock, R.W., Munk, M.M., Danehy, P.M., (...), Banks, M.E. (2026) Firefly Blue Ghost Mission 1: SCALPSS 1.1 Payload and Mission Overview. *AIAA Science and Technology Forum and Exposition, AIAA SciTech Forum 2026*. https://doi.org/10.2514/6.2026-1308

39. Viquerat, A., Schenk, M. (2016) Viscoelastic effects in metal-polymer laminate inflatable structures. *3rd AIAA Spacecraft Structures Conference*. https://doi.org/10.2514/6.2016-1221

40. Portmann, V.P., Miller, A.S., Zhang, J.Z., (...), Chin, H.H. (2024) Modeling and Feedback Control for a Guyed, Flexible, Tubular Lunar Tower. *IEEE Aerospace Conference Proceedings*. https://doi.org/10.1109/AERO58975.2024.10521355

41. Benton, M.G., Donahue, B., Caplin, G., (...), Reiley, K. (2009) Configuration options to maximize lunar surface reuse of Altair lander structure and systems. *AIAA Space 2009 Conference and Exposition*. https://doi.org/10.2514/6.2009-6405

42. Wu, K.C., Antol, J., Watson, J.J., (...), Mazanek, D.D. (2007) Lunar lander structural design studies at NASA Langley. *A Collection of Technical Papers - AIAA Space 2007 Conference*. https://doi.org/10.2514/6.2007-6137

43. Cadogan, D., Stein, J., Grahne, M. (1999) Inflatable composite habitat structures for lunar and mars exploration. *Acta Astronautica*. https://doi.org/10.1016/S0094-5765(99)00103-4

44. Cadogan, D.P., Scarborough, S.E. (2001) Rigidizable materials for use in gossamer space inflatable structures. *19th AIAA Applied Aerodynamics Conference*. https://doi.org/10.2514/6.2001-1417

45. Papike, J.J., Simon, S.B., Laul, J.C. (1982) The lunar regolith: Chemistry, mineralogy, and petrology. *Reviews of Geophysics*. https://doi.org/10.1029/RG020i004p00761

46. Williams, H., Crum, R., Ness, R.V., (...), Hell, K. (2023) Lunar Array, Mast, and Power System (LAMPS) for Deployable Lunar Power Provision. *AIAA SciTech Forum and Exposition, 2023*. https://doi.org/10.2514/6.2023-1389

47. Zhang, Z., Guo, X., Pan, B., (...), Jiang, S. (2023) Experimental study and numerical simulation of load-bearing/coiling characteristics of thin-walled lenticular composite booms. *Composite Structures*. https://doi.org/10.1016/j.compstruct.2023.117515

48. Takao, Y., Mori, O., Kawaguchi, J. (2021) Analysis and design of a spacecraft docking system using a deployable boom. *Acta Astronautica*. https://doi.org/10.1016/j.actaastro.2020.10.031

49. Barbera, D., Laurenzi, S. (2015) Nonlinear buckling and folding analysis of a storable tubular ultrathin boom for nanosatellites. *Composite Structures*. https://doi.org/10.1016/j.compstruct.2015.05.024

50. Kim, K., Royer, F. (2025) A data-driven approach for imperfection-insensitive thin-shell structure design via localized dimple imperfections and gradient boosting. *International Journal of Solids and Structures*. https://doi.org/10.1016/j.ijsolstr.2025.113637

51. Oberbeck, V.R., Quaide, W.L. (1968) Genetic implications of Lunar regolith thickness variations. *Icarus*. https://doi.org/10.1016/0019-1035(68)90039-0

52. Firth, J.A., Adamcik, B.D., Hannah, E.C., (...), Firth, D.H. (2019) Minimal unpowered strain-energy deployment mechanism for rollable spacecraft booms. *AIAA Scitech 2019 Forum*. https://doi.org/10.2514/6.2019-1258

53. Wei, X., Lin, Q., Nie, H., (...), Ren, J. (2014) Investigation on soft-landing dynamics of four-legged lunar lander. *Acta Astronautica*. https://doi.org/10.1016/j.actaastro.2014.04.001

54. Weiss, P., Yung, K.L. (2010) Feasibility study of a lunar landing area navigation network deployed by impacting micro-probes. *Planetary and Space Science*. https://doi.org/10.1016/j.pss.2010.02.007

55. Mu, Rongjun, Li, Yuntian, Luo, Rubin, (...), Shan, Yongzhi (2020) A Distributed Radio Beacon/IMU/Altimeter Integrated Localization Scheme with Uncertain Initial Beacon Locations for Lunar Pinpoint Landing. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s20195643

56. Grebow, D.J., Ozimek, M.T., Howell, K.C., Folta, D.C. (2006) Multi-body orbit architectures for lunar south pole coverage. *Advances in the Astronautical Sciences*. https://www.scopus.com/pages/publications/34247141008

57. Lu, Y., Yang, Y., Li, X., (...), Dong, G. (2012) Lunar polar orbit relay constellation and its deployment and maintenance. *Proceedings - 2012 International Conference on Intelligent Systems Design and Engineering Applications, ISDEA 2012*. https://doi.org/10.1109/ISdea.2012.448

58. Zhang, L. (2023) From Queqiao to Queqiao-2: The sustainable development of Chinese lunar relay communications satellite. *Proceedings of the International Astronautical Congress, IAC*. https://www.scopus.com/pages/publications/85188011524

59. Kerslake, T.W. (2008) Lunar surface-to-surface power transfer. *AIP Conference Proceedings*. https://doi.org/10.1063/1.2845004

60. Vigneron, Frank R., Millar, R.A. (1977) PLAN FOR FLIGHT EVALUATION OF ATTITUDE STABILIZATION AND FLEXIBLE SOLAR ARRAY DYNAMICS FOR THE COMMUNICATIONS TECHNOLOGY SATELLITE. *Space and Energy*. https://doi.org/10.1016/B978-0-08-021053-7.50010-0

61. Li, Y., Wang, Z., Wang, C., Huang, W. (2017) Effects of torque spring, CCL and latch mechanism on dynamic response of planar solar arrays with multiple clearance joints. *Acta Astronautica*. https://doi.org/10.1016/j.actaastro.2016.12.032

62. Li, Yuanyuan, Wang, Zilu, Wang, Cong, Huang, Wenhu (2016) Dynamic responses of space solar arrays considering joint clearance and structural flexibility. *Advances in Mechanical Engineering*. https://doi.org/10.1177/1687814016657945

63. Schreiner, S.S., Setterfield, T.P., Roberson, D.R., (...), Head, J.W. (2015) An overnight habitat for expanding lunar surface exploration. *Acta Astronautica*. https://doi.org/10.1016/j.actaastro.2015.03.012

64. Ma, X., Li, T., Ma, J., (...), Li, Y. (2022) Recent Advances in Space-Deployable Structures in China. *Engineering*. https://doi.org/10.1016/j.eng.2022.04.013

65. He, Y., Xu, S., Lu, G., (...), Tse, K.M. (2026) Structural design strategies for energy-absorbing auxetic metamaterials: a review. *International Journal of Mechanical Sciences*. https://doi.org/10.1016/j.ijmecsci.2026.112109

66. Klosky, J.Ledlie, Sture, Stein, Ko, Hon-Yim, Barnes, Frank (1998) Helical anchors for combined anchoring and soil testing in lunar operations. *Proceedings of the International Conference and Exposition on Engineering, Construction, and Operations in Space*. https://www.scopus.com/pages/publications/0031649830